SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1706**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Rekha S K (192511023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Rekha S K

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit

assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

i. What are the percept's for this agent? ii. Characterize the operating environment. iii. What are the actions the agent can take? iv. How can one evaluate the performance of the agent?

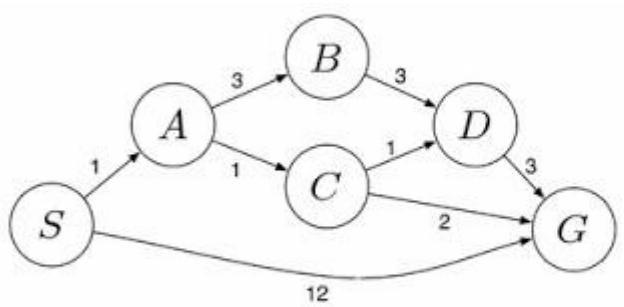
v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ANSWERS

1. WATER JUG PROBLEM

Initial State

- 4-gallon jug = 0 gallons
- 3-gallon jug = 0 gallons

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour the 3-gallon jug into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour from the 3-gallon jug into the 4-gallon jug until it is full	4	2
5	Empty the 4-gallon jug onto the ground	0	2
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	**2**	**0**

Final State

- 4-gallon jug = 2 gallons
- 3-gallon jug = 0 gallons

State Representation

(0,0)
↓ Fill 3-gallon jug
(0,3)
↓ Pour 3 → 4
(3,0)
↓ Fill 3-gallon jug
(3,3)
↓ Pour 3 → 4 until full
(4,2)
↓ Empty 4-gallon jug
(0,2)
↓ Pour remaining 2 gallons into 4-gallon jug
(2,0) ← Goal State

The **4-gallon jug contains exactly 2 gallons**.

2. MARS ROVER – ANALYSIS AND EXPLORATION OF SAMPLES ON MARS

i. What are the percepts for this agent?

Percepts are the information received by the Mars Rover through its sensors.

- Images from cameras
- Rock and soil sample data
- Temperature readings
- Atmospheric pressure
- Chemical composition of rocks and soil
- Terrain and obstacle information
- Position and direction (GPS-like navigation system)

ii. Characterize the operating environment.

The Mars Rover operates in the following environment:

- **Partially Observable:** The rover cannot observe the entire environment at once.
- **Deterministic:** Actions usually produce predictable results.
- **Sequential:** Current actions affect future actions.
- **Dynamic:** Weather conditions such as dust storms can change the environment.
- **Continuous:** Movement, time, and location are continuous.
- **Single Agent:** The rover performs tasks independently.

iii. What are the actions the agent can take?

The Mars Rover can perform the following actions:

- Move forward, backward, left, or right
- Capture images
- Collect rock and soil samples
- Analyze samples using onboard instruments
- Avoid obstacles
- Send collected data to Earth
- Drill into rocks
- Stop or change direction when required

iv. How can one evaluate the performance of the agent?

The performance of the Mars Rover can be evaluated based on:

- Successful collection of rock and soil samples
- Accuracy of scientific analysis
- Safe navigation without collisions
- Efficient use of battery power
- Amount of useful scientific data transmitted to Earth
- Completion of assigned missions within the planned time

v. What sort of agent architecture do you think is most suitable for this agent? Why?

Most Suitable Agent Architecture: Utility-Based Agent (or **Learning Agent**)

Reason:

- It selects the best action by considering safety, battery level, and mission goals.

- It can make intelligent decisions in an unknown environment.
- It maximizes scientific discoveries while minimizing risks.
- It can improve its performance based on previous experiences and collected data.

3. 8-QUEENS PROBLEM

i. Initial State

- An **empty 8 × 8 chessboard** with no queens placed.

ii. State Space

- All possible arrangements of placing 8 queens on the chessboard.
- In AI, one queen is placed in each column, reducing the search space.

iii. Actions (Operators)

The possible actions are:

- Place a queen in any safe row of the next column.
- If no safe position exists, backtrack and try another position.

iv. Goal Test

A solution is reached when:

- All **8 queens** are placed on the board.
- No two queens share:
    - the same row,
    - the same column,
    - the same diagonal.

v. Path Cost

- Each queen placement has a cost of **1**.
- Since every solution places exactly 8 queens, all valid solutions have the same total cost.
- Therefore, the objective is to find **any valid solution**, not the minimum-cost one.

Search Strategy

The most suitable search strategy is **Backtracking Search (Depth-First Search)**.

Steps:

1. Place a queen in the first column.
2. Move to the next column and place a queen in a safe position.
3. Continue placing queens column by column.
4. If no safe position is available, backtrack to the previous column and try another row.
5. Repeat until all 8 queens are placed successfully.

```
[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]
```

Here 1 represents the Queen

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

4. OLA CAB BOOKING PROBLEM

i. Type of Problem-Solving Agent
Problem-Solving Agent (Goal-Based Agent)
Reason:

- The customer's goal is to reach the destination.
- The agent considers available cab options and selects the most suitable one based on the customer's preference (cost, comfort, and availability).
- It plans the best sequence of actions to achieve the goal.

ii. Pseudocode

```
START

Input source location
Input destination location

Display available cab types
(Mini, Micro, Sedan, Shared, Prime)

Customer selects a cab type
IF selected cab is available THEN
    Assign the nearest driver
    Calculate fare
    Confirm booking
    Display driver details
    Track cab until destination
ELSE
    Display "Selected cab is not available"
    Ask customer to choose another cab
ENDIF
END
```

iii. Flow of the Agent

```
Start
  ↓
Enter Source and Destination
  ↓
Display Available Cab Types
  ↓
Customer Selects Cab
  ↓
Is Cab Available?
  ↓
Yes -------------------- No
↓                         ↓
Assign Driver          Show "Cab Not Available"
↓                         ↓
Calculate Fare         Choose Another Cab
↓
Confirm Booking
↓
Track Ride
↓
Reach Destination
↓
End
```

iv. Performance Measure

The agent performs well if it:

- Finds the nearest available cab.
- Minimizes waiting time.
- Chooses the best route.
- Calculates the correct fare.
- Ensures the customer reaches the destination safely.
- Completes the booking successfully.

5. UNIFORM COST SEARCH (UCS)

Graph

Edges and Costs:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$

- C → G = 2
- D → G = 3

UCS Search Process

Step	Expanded Node	Path Cost	Priority Queue
1	S	0	A(1), G(12)
2	A	1	C(2), B(4), G(12)
3	C	2	G(4), D(3), B(4), G(12)
4	D	3	G(4), B(4), G(6), G(12)
5	G	**4**	Goal Reached

Least-Cost Path
Possible paths are:

1. **S → G = 12**
2. **S → A → B → D → G = 1 + 3 + 3 + 3 = 10**
3. **S → A → C → D → G = 1 + 1 + 1 + 3 = 6**
4. **S → A → C → G = 1 + 1 + 2 = 4** ☑

Optimal Solution
Least-Cost Path:
S → A → C → G
Total Cost:
1 + 1 + 2 = 4

UCS Algorithm (Pseudocode):
Uniform_Cost_Search(Start, Goal)

1. Insert Start into the priority queue with cost 0.
2. Repeat until the queue is empty:
    a. Remove the node with the lowest path cost.
    b. If the node is Goal, return the path and cost.
    c. Expand all neighboring nodes.
    d. Add neighbors with updated cumulative cost.
3. If Goal is not found, return Failure.

End